Dr Sofia Almpani

Postdoctoral researcher, Adjunct university professor

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Research interests and vision

My research focuses on **Trustworthy and Human-Centred AI**, integrating computational argumentation with responsible AI principles to develop intelligent systems that are transparent, accountable, and responsive to human needs. My future research follows three interconnected directions:

- **Computational argumentation and formal reasoning.** Developing formal frameworks for explainable, verifiable, and traceable reasoning in intelligent and multi-agent system.
- **Human-centered and accountable AI.** Advancing accountable AI systems that support human oversight, transparent decision-making, ethical reasoning, and regulatory compliance.
- **Applications of intelligent systems.** Applying trustworthy AI approaches to cybersecurity, privacy and safety-critical systems, including aerospace, wearable robotics and medical devices.

My vision is to advance interdisciplinary research, strengthen international collaborations, pursue competitive funding, and engage students in research connecting AI theory with real-world challenges.

Research and professional experience

Oct 2022–present — **Postdoctoral Researcher, National Technical University of Athens, Greece**

Papakuriakopoulos scholarship "Mathematics of Computer Science" (2024)

- Conduct research in argumentation logic and explainable AI, with applications to compliance, requirements engineering, and human-governed decision-making systems.
- Mentor doctoral and early-career researchers
- Publish collaborative research and contribute to research-proposal writing.

Apr 2025–present — **Researcher, Institute of Digital Innovation and Research (IDIR), Ireland**

- Contribute to the Horizon Europe AVALANCHE project.
- Publish collaborative research and contribute to research-proposal writing.

Apr 2021–present — **Researcher, ENGINO Net Ltd, Educational R&D Department, Cyprus**

Research and curriculum development at the interface of robotics, STEM pedagogy, and product innovation, including contributions to the GinoBot and MakerSpace projects.

- Prepare and manage research proposals on robotics, maker education, and digital STEM learning; Propose new educational products and research areas.
- Coordinated and mentored the Educational R&D team, assigning tasks and responsibilities, providing ongoing guidance and feedback, and supporting the development of STEM curricula and educational materials.
- Develop original, age-appropriate K–12 STEM lesson plans combining engineering design thinking, problem-based learning, hands-on experimentation, and supporting theory.
- Create teacher manuals, curriculum mappings, and design and manage digital learning resources using ENGINO LMS and Moodle.

Jan 2019–Jun 2021 — **Researcher, Edumotiva, European Lab for Educational Technology, Greece**

- Undertook research for the Horizon 2020 INBOTS project, focusing on robotics in education and societal uptake.
- Contributed to project deliverables, dissemination, and workshop organization.

Higher Education Teaching

My teaching combines conceptual foundations with hands-on problem-solving in programming, algorithms, databases, machine learning, computational mathematics and logic. I design and deliver lectures, laboratory exercises, online learning materials, and assessments, using practical examples and constructive feedback to support students with diverse backgrounds.

Undergraduate teaching

Sep 2025–present — **University of Athens — Adjunct professor, Secondment**
Department of Aerospace Science and Technology
- Database (SQL)
- Algorithms, Programming and Data Structures (C)
- Data Mining and Machine Learning (SciKit-Learn, Keras)
- Space and Cybersecurity (Python)

Mar 2026– present — **Hellenic Air Force Academy — Teaching Personnel under Contract Status**
- Big Data (Spark, Hadoop, MapReduce, Kafka, MongoDB)
- Cloud Computing

Sep 2025– Feb 2026 — **University of Thessaly — Adjunct professor**
Department of Mathematics
- Topics in Computational and Applied Mathematics I

Sep 2023–Sep 2024 — **University of Athens — Secondment from Public Education**
Department of Technologies of Digital Industry
- Introduction to Programming (C)
- Machine Learning
- Object-oriented Programming (C++)
- Python

Sep 2018–Feb 2021 — **National Technical University of Athens — Teaching Assistant**
School of Applied Mathematical and Physical Sciences
- Introduction to Object-oriented Programming (Java)

Postgraduate teaching

Feb 2026–present — **MSc in Logic (NTUA, UoA & Academy of Athens),**
- Topics in Logic II

Feb 2026–present — **Neapolis University Pafos — Visiting Lecturer**
MSc in Data Analytics and Financial Technology,
- Disruptive Technologies

Sep 2026–present
MSc in Information Systems and Digital Innovation
- Disruptive Technologies and Digital Transformation

Mar 2026 and Mar 2020 — **MSc Applied Biomechanics in Orthopaedics** (Medical School, UoA)
- Wearable Robotics and Exoskeletons in Healthcare (2 lectures)

Mentoring student research teams

Feb 2026–present — Mentor to the **Perseus team (NKUA)** for 1st National Security Challenge
- Youth Entrepreneurship Award

Sep 2026–present — Mentor to the **Ikaros team (NKUA)** for the EuroBot robotic competition

Additional teaching experience

Sep 2009–present — **Teacher, Greek public primary schools**

Sep 2012–May 2022 — **Dance teacher and choreographer** (ISTD Advanced 2 and DDE Diplomas)

Education

Oct 2017–Apr 2022 — **PhD, School of Electrical and Computer Engineering**
National Technical University of Athens

Thesis: *Argumentation and Rule-based Logic in Mathematical Proving and Legal Artificial Intelligence Applications.*

PhD courses:
- Artificial Neural Networks and Machine Learning
- Advanced Topics in Data Science and Analytics
- Advanced Topics in Artificial Intelligence
- Self-Organizing (Ad Hoc) Networks
- Mobile and Personal Communications
- Multiscale Simulation of Cancer and In Silico Medicine

Jun 2014–Jul 2015 — **MSc, Educational Leadership, Management and Emerging Technologies**

Università degli Studi Guglielmo Marconi | Grade: 9/10

Thesis: Teaching of Mathematics in Higher Education and E-learning.

Oct 2009–Mar 2014 — **BSc, Department of Mathematics, University of Athens**

Direction*: Pure Mathematics* | Grade: 6.97/10
Specialization*: Teaching of mathematics*

Oct 2005–Jul 2009 — **BSc, Faculty of Education, Aristotle University of Thessaloniki**

Primary Education | *Grade: 8.79/10*

Technical skills and languages

Programming. Python, Java, C/C++, SQL, Prolog, RuleML; AI and Machine Learning (Scikit-Learn, Keras, neural networks), Big Data (SPARK, Hadoop, MongoDB, Kafka)

Digital teaching. Moodle e-learning environment certification (KE.PLH.NE.T.); Cambridge International Diploma in IT Skills; Microsoft Office.

Languages. Greek (native); English (University of Michigan Proficiency).

Honours and distinctions

- 2026 — **Youth Entrepreneurship Award, Perseus team (UoA)** for the proposal "*ASPIDA: Autonomous Swarm for Protection of Critical Infrastructures with Anti-UAV*", 1st National Security and Innovation Challenge organized by National Intelligence Service (EYP)
- 2025 — **Best Paper Award,** "Development of a Logical Reasoning System for Ensuring Compliance with the Data Protection Regulation".
- 2024 — **Papakuriakopoulos Postodoctoral scholarship** "Mathematics of Computer Science", NTUA
- 2018 — **Rule Challenge Award**, 12th International Rule Challenge, RuleML+RR, Luxembourg, for "Formalizing Air Traffic Control Regulations in PSOA RuleML".
- 2019 — **Participation grant**, ACAI/HAISS Summer School, from EETN (1–5 Jun 2019).
- 2018 — **Young Researcher Participation Grant**, LuxLogAI, Luxembourg (Sep 2018).
- 2010 — **Eurobank scholarship** "The Great Moment for Education", for first ranking in the Panhellenic entrance school examinations.
- 2009 — **I.K.Y. scholarship**, second ranking in the Faculty of Education, School of Primary Education.

Research projects

Apr 2025– Oct 2026

Consortium:
1. UBITECH – Cyprus
2. Netcompany
3. ITML – Cyprus
4. IDIR – Ireland
5. Quadible
6. Serviciul de Protecție și Pază SPP – Romania
7. Arthur’s Legal – Netherlands

AVALANCHE *Horizon 2021-2027 - IDIR*

Grant / funding. Horizon Europe GA 101168393, **Budget**: €1,499,800
Project focus. Detection of disinformation, hate speech, and deepfakes across social media and the web using behavioural and sentiment analysis.
Role and contribution: Investigated analysis of hostile online narratives, including socio-technical requirements, training curriculum development, and dissemination.
Publications:
– *Requirements Engineering for Countering Hostile Narratives: A Vision for Socio-technical Intervention Systems beyond Detection*
– *Requirements Modelling for Hate Speech Detection and Analysis*
– *Archetypical Modelling for Cybersecurity and Personal Data Protection for SMEs*

Dec 2022– Dec 2024

Consortium:
1. ENGINO – Cyprus
2. University of Cyprus

MakerSpace *Innovate - Engino Net Ltd*

Grant / funding. RIF INNOVATE/1221/0052, **Budget**: €2,112,000
Project focus. Pedagogical approaches to STEM and the engineering design process through CAD, 3D printing, laser cutting, and maker technologies.
Role and contribution. Contributed to writing the funded project proposal and to educational R&D associated with the maker-education approach.
Publication:
– *Analysing Students’ Mathematics Performance through the Use of Construction Toys*

Dec 2020– May 2023

Consortium:
1. Warsaw University of Technology – Poland
2. EDUMOTIVA – Greece
3. CYRIC – Cyprus
4. NTUA – Greece
5. Heron – Cyprus
6. University of Florence
7. Leonardo da Vinci – Italy
8. Liceum gólnokształcące,– Poland

STEAM2GO *Erasmus+ - NTUA*

Grant / funding. Erasmus-2020-1-PL01-KA201-081698, **Budget**: €229,845
Project focus. Mobile makerspaces designed to broaden access to STEAM and maker education in schools.
Outcomes: Replicable mobile makerspace model, the STEAM2GO educational backpack, a digital learning platform, and educational materials and teacher-training resources
Role and contribution. Contributed to writing the funded proposal and, in the development of the Moodle digital platform and in the educational material for mobile makerspaces.

Apr 2021– Nov 2022

Consortium:
1. ENGINO – Cyprus
2. University of Cyprus

GinoBot *RESTART 2016–2020 programme - Engino Net Ltd*

Grant / funding. RIF INNOVATE/0719/0098, **Budget**: €1,603,692.40
Project focus. Development of an expandable robot and supporting product ecosystem for STEM education.
Role and contribution. Researcher developing the curriculum and educational materials associated with the GinoBot learning ecosystem.

Jan 2019– Jun 2021

Consortium – 25 partners including: Spanish National Research Council, **EDUMOTIVA**, TECNALIA, Complutense University of Madrid, Vrije Universiteit Brussel, University of Siena, Ottobock, KTH Royal Institute of Technology, University of Leeds, Utrecht University, Space Applications Services, PAL Robotics, and others.

INBOTS *Horizon 2020 - Edumotiva*

Grant / funding. Horizon 2020 GA 780073 · **Budget**: €2,982,973.75
Project focus. Inclusive robotics and the societal uptake of robotics.
Role and contribution. Contributed to the education and societal-uptake work, and helped organize project workshops.
Outputs and dissemination. Co-authored the deliverable “Robotics in Education and the Societal and Socio-economic Uptake of Robotics” (WP6, with D. Alimisis); contributed to a project interview with Engino founder Costas Sisamos.
Publication:
– *Dance and Robots: Designing a Robotics-Enhanced Project for Dance-Based STEAM Education Using ENGINO*

Jun 2020–Feb 2021

Consortium:
1. Association des Professionnels des Industries de la Langue (APIL) – France
2. NTUA – Greece
3. Future Now – Greece

LegiCrowd *Next Generation Internet / Horizon 2020 - NTUA*

Grant / funding. NGI TRUST Call 2, project 2.31
Project focus. Ontology for AI-supported annotations of online terms of service, aimed at greater transparency and data privacy.
Role and contribution. Contributed to writing the funded NGI TRUST subproject proposal and to the subsequent ontology-oriented research and development work.
Publication: – From Legal Documents to Legal Document Management Systems: The Case of LegiCrowd

Mar 2018–Dec 2020

Wearable Robots *COST Action 16116 - NTUA*

Project focus. Wearable robots for the augmentation, assistance, or substitution of human motor functions.
Outputs and dissemination. Short-Term Scientific Mission «*Effective education of engineers towards the ethical, legal and societal aspects of wearable robotics*» University of Twente, the Netherlands (28 Aug–3 Sep 2019); work on material addressing the ethical, legal and societal aspects of wearable robotics.
International engagement. Network meetings in Darmstadt (5–6 Mar 2018), Enschede (Aug 2018), Bucharest (18–19 Mar 2019), and Berlin (22–23 Oct 2019).

- **Publications:** ExosCE: A Legal-based Computational System for Compliance with Exoskeletons CE Marking
- Converging Robotic Technologies in Targeted Neural Rehabilitation: A Review of Emerging Solutions and Challenges.
- Neurorobotics: Review of Underlying Technologies, Current Developments, and Future Directions
- Formalization of Ethical Decision Making: Implementation in the Data Privacy of Wearable Robots
- Supporting Ethical Decisions in Wearable Technology with Deontic Logic

Research Proposal writing

Actively contributed to **over 20 research and innovation proposals** across Horizon Europe, Erasmus+, and national funding programmes, covering AI, cybersecurity, robotics, STEAM education, and digital innovation. Responsibilities included **proposal writing, budget planning, consortium building, partner coordination, and electronic submission**. In addition to the funded submissions listed above, selected unfunded proposals are presented below.

- **DATASPARK** | Horizon Europe, HORIZON-CL4-2026-04-DATA-06 (2026) - IDIR
  Contributed to Part B, Section 1 (Excellence), addressing interoperable AI-ready data pipelines, compliance, governance, and research methodology.
- **CYBERinSMEs** | Erasmus+ Capacity Building in Higher Education, Strand 2 (2025) - NTUA
  Contributed to the work packages on developing cybersecurity educational modules and micro-credentials for SMEs in Armenia and Montenegro.
- **STEAM Exploratorium** | Research Excellence Partnerships (ΣΕΑ), Action 16289 (2024) - Engino
  Authored the complete proposal for a travelling interactive STEAM exhibition, including the funding-related work package, and managed the online submission and documentation.
  - Passed the evaluation threshold but was not funded due to budget limitations**.**
- **HERACLITUS – Human-centred Ethical Reasoning and AI** | Fellowship in AI Ethics, Center for Hellenic Studies, Harvard University (2024) - NTUA
  Authored and submitted the full proposal for an interdisciplinary AI ethics curriculum

Scholarly Activities

Conference organization

21–23 Oct 2026 **17th Conference on Scalable Uncertainty Management (SUM 2026), Athens**
Organizing Committee

1–3 Dec 2026 **19th Conference on the Practice of Enterprise Modeling (PoEM 2026), Mytilene**
Scientific Committee

25-27 Sep 2026 **2nd Interdisciplinary Conference of PhD Holders of the Public Domain, Athens**
26-28 Sep 2025 **1st Interdisciplinary Conference of PhD Holders of the Public Domain, Athens**
Scientific and Organizing Committees

23–25 Sep 2022 **3rd Panhellenic Scientix Conference, Athens**
Scientific and Organizing Committees

1–11 Apr 2022 **7th World Congress and School on Universal Logic (UNILOG), Trends in Argumentation Logic Workshop, Crete**
Scientific Committee

19-20 Dec 2019 **Rules: Logic and Applications Workshop 2019, NTUA, Athens**
16-17 Dec 2018 **Rules: Logic and Applications Workshop 2018, NTUA, Athens**
Scientific and Organizing Committees

Editorial contributions

- Editorial Board, **Studia Humana**.
- Co-editor with Schumann, A., Lisanyuk, E, **Trends in Argumentation Logic**, Studia Humana, 2022

Selected participation in international networks and related publications

- **PSOA RuleML and PSOATransRun Group,** Harold B., University of New Brunswick, Canada
  - *Development of a Logical Reasoning System for Ensuring Compliance with the Data Protection Regulation*
  - *Aligning, Interoperating, and Co-executing Air Traffic Control Rules across PSOA RuleML and IDP*
  - *Formalizing Air Traffic Control Regulations in PSOA RuleML*
  - *A Rule-Based Model for Compliance of Medical Devices Applied to the European Market*
  - *Computational Regulation of Medical Devices in PSOA RuleML*

- **Hellenic-Romanian Logic and Computation Group**
  - *Bridging Informal Reasoning and Formal Proving: The Role of Argumentation in Proof-Events*
  - *Formalization of Mathematical Proof Practice Through an Argumentation-Based Model*

- **Jadavpur University, Kolkata, India,** Chakraborty M.
  - *A Dualistic Interpretation of Mathematical Creation through Art and Argumentation*
  - *Integrating Argumentation, Insights, and Epistemic Logic for a Universal Perspective on Computational Creativity*

- **University of Twente, The Netherlands,** Veneman, J., Nizamis, K
  - *Converging Robotic Technologies in Targeted Neural Rehabilitation: A Review of Emerging Solutions & Challenges*
  - *Neurorobotics: Review of Underlying Technologies, Current Developments, and Future Directions*

- **λ-form Formal Methods Research Group**
  - *Exploring the Role of Argumentation in Human-Centered Artificial Intelligence: Communication, Collaboration, Trust*
  - *A Collaborative Argumentation Framework for Goal-Oriented Reasoning.*

- **Scientix, European educational collaboration**, Ambassador for Greece.
  - *Augmented Reality Escape Room in STEM: Newton's Kidnap*

Complete list of publications

Google Scholar (Oct 2026): h-index 8; 300+ citations. Publication collection Drive

Peer-reviewed journal articles

1. **Almpani, S**., Loucopoulos, P., & Plithas, A. (2026). *Requirements Engineering for Countering Hostile Narratives: A Vision for Socio-technical Intervention Systems beyond Detection.* Requirements Engineering, 31, article 18. https://doi.org/10.1007/s00766-026-00473-w

2. **Almpani, S.,** & Stefaneas, P. (2026). *Exploring the Role of Argumentation in Human-Centered Artificial Intelligence: Communication, Collaboration, and Trust.* International Journal on Artificial Intelligence Tools. https://doi.org/10.1142/S0218213026500144

3. **Almpani, S**., Gantzounis, A., & Stefaneas, P. (2026). *A Collaborative Argumentation Framework for Goal-Oriented Reasoning.* Requirements Engineering, 31, article 8. https://doi.org/10.1007/s00766-026-00461-0

4. Sofroniou, A., Premnath, B., Sisamos, C., **Almpani, S.,** & Sisamou, E. (2025). *Analysing Pre-primary Students' Mathematics Performance through the Use of Educational Construction Toys.* Journal of Mathematics, Science and Technology Education, 2(1), 19–32. https://doi.org/10.12973/jmste.2.1.19

5. **Almpani, S.,** Stefaneas, P., & Chakraborty, M. (2025). *A Dualistic Interpretation of Mathematical Creation through Art and Argumentation.* Journal of Humanistic Mathematics, 15(2), 237–273. https://scholarship.claremont.edu/jhm/vol15/iss2/11

6. **Almpani, S**., & Stefaneas, P. (2023). *Bridging Informal Reasoning and Formal Proving: The Role of Argumentation in Proof-Events.* Foundations of Science, 30, 201–225. https://doi.org/10.1007/s10699-023-09926-9

7. **Almpani, S**., Stefaneas, P., & Frangos, P. (2023). *Formalization of Ethical Decision Making: Implementation in the Data Privacy of Wearable Robots.* International Journal of Extreme Automation and Connectivity in Healthcare, 5(1), 1–13. https://doi.org/10.4018/IJEACH.320488

8. **Almpani, S**., Stefaneas, P., & Vandoulakis, I. (2023). *Formalization of Mathematical Proof Practice Through an Argumentation-Based Model.* Axiomathes, 33(3), 1–28. https://doi.org/10.1007/s10516-023-09685-z

9. Zafeirakopoulos, D., **Almpani, S**., & Stefaneas, P. (2022). *Supporting Ethical Decisions in Wearable Technology with Deontic Logic: A Brief Introduction.* Electronic Communications of the EASST, 81. https://doi.org/10.14279/tuj.eceasst.81.1192

10. **Almpani, S**., Stefaneas, P., & Frangos, P. (2022). *Argumentation-Based Logic for Ethical Decision Making.* Studia Humana, 11(3–4), 46–52. https://doi.org/10.2478/sh-2022-0015

11. **Almpani, S.,** Kiouvrekis, Y., Stefaneas, P., & Frangos, P. (2022). *Computational Argumentation for Medical Device Regulatory Classification.* International Journal on Artificial Intelligence Tools, 31(1). https://doi.org/10.1142/S0218213022500051

12. Nizamis, K., Athanasiou, A., **Almpani, S**., Dimitrousis, C., & Astaras, A. (2021). *Converging Robotic Technologies in Targeted Neural Rehabilitation: A Review of Emerging Solutions and Challenges.* Sensors, 21(6), 2084. https://doi.org/10.3390/s21062084

13. **Almpani, S.,** Mitsikas, T., Stefaneas, P., & Frangos, P. (2020). *ExosCE: A Legal-based Computational System for Compliance with Exoskeletons CE Marking.* Paladyn, Journal of Behavioral Robotics, 11(1), 414–427. https://doi.org/10.1515/pjbr-2020-0026

14. **Almpani, S.,** Stefaneas, P., Boley, H., Mitsikas, T., & Frangos, P. (2019). *A Rule-Based Model for Compliance of Medical Devices Applied to the European Market.* International Journal of Extreme Automation and Connectivity in Healthcare, 1(2), 56–78. https://doi.org/10.4018/IJEACH.2019070104

Accepted manuscripts

1. **Almpani, S.,** Stefaneas, P., & Chakraborty, M. (2026). *Integrating Argumentation, Insights, and Epistemic Logic for a Universal Perspective on Computational Creativity.* Logica Universalis. (under publication)

Book chapter and editorial

1. Dimitrousis, C., **Almpani, S**., Stefaneas, P., Veneman, J., Nizamis, K., & Astaras, A. (2020). *Neurorobotics: Review of Underlying Technologies, Current Developments, and Future Directions.* In Neurotechnology: Methods, Advances and Applications, 181–213. https://doi.org/10.1049/PBHE019E_ch7

2. Schumann, A., Lisanyuk, E., & **Almpani, S.** (2022). *Trends in Argumentation Logic.* Studia Humana, 11(3–4), 1–5. https://doi.org/10.2478/sh-2022-0010

Peer-reviewed conference publications

1. **Almpani, S**. (2026). *APEC-PS: An Argumentation Framework for Safe and Traceable Agentic AI.* 2nd International Interdisciplinary Conference of PhD Holders of the Public Domain, Athens. (under publication)

2. **Almpani, S.** (2026). *Temporal Argumentation for Continuous Cyber-Safety Assurance of AI-Enabled Aerospace Systems.* SUM 2026, Athens, 21–23 Oct 2026. (under publication)

3. **Almpani, S.,** Gantzounis, A., & Stefaneas, P. (2026). *Argumentation Logic for Explainable Goal-Oriented Reasoning in Multi-agent Systems.* 3rd Congress of Greek Mathematicians, Athens, 29 Jun–4 Jul 2026. http://www.math.ntua.gr/~vgregoriades/TCGM_2026.html

4. **Almpani, S**., Loucopoulos, P., & Skourtis, Y. (2026). *Archetypical Modelling for Cybersecurity and Personal Data Protection for SMEs.* 17th EAI Digital Forensics and Cyber Crime, Reykjavik, 8–10 Sep 2026.

5. **Almpani, S.,** Kavakli, E., Loucopoulos, P., & Valari, E. (2026). *Requirements Modelling for Hate Speech Detection and Analysis.* WorldCist 2026, Madeira, 31 Mar–2 Apr 2026. https://doi.org/10.1007/978-3-032-32032-2_1

6. **Almpani, S.,** Stefaneas, P., & Boley, H. (2025). *Development of a Logical Reasoning System for Ensuring Compliance with the Data Protection Regulation.* 1st International Interdisciplinary Conference of PhD Holders of the Public Domain, Athens, Sep 2025. pasyd.gr/ ΒΙΒΛΙΟ-ΠΡΑΚΤΙΚΩΝ_ΤΟΜΟΣ-Β.pdf

7. Sofroniou, A., Premnath, B., Sisamos, C., **Almpani, S.,** & Sisamou, E. (2024). *Analysing Student Performance in Mathematics of Pre-primary Students through Engino Toy Blocks.* 6th Global Conference on Education and Teaching, France, 25–27 Oct 2024. https://www.dpublication.com/wp-content/uploads/2024/10/29-4109.pdf

8. Tsantali, E., **Almpani, S.,** & Atzampou, G. (2024). *Exploring Collective Memory and Trauma through a Site-Specific Performance at Lazaretta.* Performing Space 2024, Greece. https://eproceedings.epublishing.ekt.gr/index.php/ps/article/view/8366/7564

9. **Almpani, S**., Tsantali, E., & Bitsicas, K. (2024). *Expressing Collective Memory through Embodied Dialogues and Audiovisual Design.* Digital Culture & AudioVisual Challenges, Corfu/online, 24–25 May 2024.

10. Marinou, E., **Almpani, S**., Sisamou, E., & Sisamos, C. (2022). *Augmented Reality Escape Room in STEM: Newton's Kidnap.* 3rd Panhellenic Scientix Conference, Athens, 23–25 Sep 2022.

11. **Almpani, S**., & Stefaneas, P. (2022). *Argumentation-based Checking of Misinformation with Formal Calculus for Truth-assignment.* Trends in Argumentation Logic Workshop, UNILOG, Chania, 6–11 Apr 2022.

12. Zafeirakopoulos, D., Almpani, S., & Stefaneas, P. (2021). *Supporting Ethical Decisions in Wearable Technology with Deontic Logic: A Brief Introduction.* 9th ISoLA Doctoral Symposium. https://doi.org/10.14279/tuj.eceasst.81.1192

13. **Almpani, S.,** Kiouvrekis, Y., & Stefaneas, P. (2021). *Modeling of Medical Devices Classification with Computational Argumentation.* IISA 2021, Chania, 12–14 Jul 2021. https://doi.org/10.1109/IISA52424.2021.9555530

14. **Almpani, S.** (2021). *E-learning and Teaching of Mathematics in Higher Education.* 2nd International Scientific Conference "Machines. Humans. Values", Russia.

15. **Almpani, S.,** & Almisis, D. (2021). *Dance and Robots: Designing a Robotics-Enhanced Project for Dance-Based STEAM Education Using ENGINO.* EDUROBOTICS 2021, Studies in Computational Intelligence 982. https://doi.org/10.1007/978-3-030-77022-8_13

16. Nousias, A., Couillault, A., **Almpani, S.,** Mitsikas, T., & Stefaneas, P. (2020). *From Legal Documents to Legal Document Management Systems: The Case of LegiCrowd.* Artificial Intelligence, Ethics and Law Workshop, SETN 2020.

17. Deryck, M., Mitsikas, T., **Almpani, S.,** Stefaneas, P., Frangos, P., Ouranos, I., Boley, H., & Vennekens, J. (2019). *Aligning, Interoperating, and Co-executing Air Traffic Control Rules across PSOA RuleML and IDP.* RuleML+RR 2019, LNCS 11784. https://doi.org/10.1007/978-3-030-31095-0_4

18. **Almpani, S.,** Stefaneas, P., & Vandoulakis, I. (2019). *On the Significance of Argumentation in Discovery Proof-Events.* 16th Congress on Logic, Methodology and Philosophy of Science and Technology, Prague, 5–10 Aug 2019.

19. **Almpani, S.,** Gantzounis, A., & Stefaneas, P. (2018). *Zero Knowledge Proof Events: Ali Baba's Cave.* ISSC 2018, Logics of Image, Kolymbari, Crete, Aug 2018. Keynote.

20. **Almpani**, **S**., Stefaneas, P., Boley, H., Mitsikas, T., & Frangos, P. (2018). *Computational Regulation of Medical Devices in PSOA RuleML.* RuleML+RR 2018, LNCS 11092. https://doi.org/10.1007/978-3-319-99906-7_13

21. **Almpani, S.,** Stefaneas, P., Boley, H., Mitsikas, T., & Frangos, P. (2018). *Object-Relational Rules for Medical Devices: Classification and Conformity.* OTM 2018, LNCS 11230. https://doi.org/10.1007/978-3-030-02671-4_37

22. Mitsikas, T., **Almpani, S.,** Stefaneas, P., Frangos, P., & Ouranos, I. (2018). *Formalizing Air Traffic Control Regulations in PSOA RuleML.* RuleML+RR 2018, Doctoral Consortium and Challenge, vol. 2204. Rule Challenge Award. https://ceur-ws.org/Vol-2204/paper9.pdf

23. **Almpani, S**., & Stefaneas, P. (2017). *E-learning and Teaching of Mathematics in Higher Education.* ICT and Education, NTUA (in Greek).

24. **Almpani, S.,** Stefaneas, P., & Vandoulakis, I. (2017). *On the Role of Argumentation in Discovery Proof-Events.* C3GI 2017, 6th International Workshop on Computational Creativity, Concept Invention and General Intelligence. https://ceur-ws.org/Vol-2160/C3GI_2017_paper_6.pdf

25. **Almpani, S**., & Stefaneas, P. (2017). *On Proving and Argumentation.* AIC 2017, 5th International Workshop on Artificial Intelligence and Cognition, Larnaca. https://ceur-ws.org/Vol-2090/paper7.pdf

Work under review

1. **Almpani, S.** (2026). *Traceable and Contestable Reasoning for Trustworthy Human-Governed Agentic AI.* ACM Transactions on Autonomous and Adaptive Systems. (Under review).

2. **Almpani**, **S.** (2026) *Human-Centered Agentic AI with Argumentation-Based Decision Support for UAV Cybersecurity.* 2nd International Conference on Human-AI Collaboration & Augmented Intelligence. (Under review).